



Measures of Disease Occurrence

APHS 20203 • Epidemiology & Biostatistics

Measurement, Uncertainty, and Study Design

MEASUREMENT

UNCERTAINTY

STUDY DESIGN

Epidemiology Begins With Occurrence and Distribution

“The study of the occurrence and distribution of health-related states or events in specified populations, including the study of the determinants influencing such states, and the application of this knowledge to control the health problems.”

Source: (Porta 2008)

Measurement Makes Characteristics Observable

MEASUREMENT

- Observe and assign values to relevant characteristics.
- Look for patterns among those values.

“The study of the occurrence and distribution of health-related states or events in specified populations, including the study of the determinants influencing such states, and the application of this knowledge to control the health problems.”

Source: (Porta 2008)

What Does It Mean to Observe?

MEASUREMENT

Observe and assign values to relevant characteristics.

OBSERVE

"To be aware of" or "have knowledge about."

Measurement Produces Data; Statistics Finds Patterns

MEASUREMENT

Observe characteristics.

DATA

X	Y	Z
1	A	1
2	B	0
3	C	1

STATISTICS

Look for patterns among the values.

Description, Prediction, and Causation

DESCRIPTION

PREDICTION

CAUSATION

Description Does Not Necessarily Seek Associations

- Description examines distributions of single variables: middle, spread, shape, count, and proportion.
- It supports resource management and planning.
- Examples:
 - How many ventilators are available in Texas?
 - What is the average age of people living in Florida?
 - How much time elapses, on average, between exposure to a pathogen and the occurrence of disease symptoms?

Description Can Also Compare Distributions

- Sometimes description does look for associations.
- It may compare distributions of single variables within levels of another variable.
- Examples:
 - Are more ventilators available in Texas or New York?
 - Are people older, on average, in Florida or Pennsylvania?
 - Is symptom onset quicker, on average, for cholera or *E. coli*?

Measures of Occurrence Support Description

- Counts
- Incidence
- Prevalence
- Odds

These measures can be useful on their own and for hypothesis generation.

Sources of Uncertainty: Few Checklists

1. Few checklists

Sources of Uncertainty: Add Statistical Uncertainty

1. Few checklists
2. Statistical uncertainty

Sources of Uncertainty: Add Causal Uncertainty

1. Few checklists
2. Statistical uncertainty
3. Causal uncertainty

Occurrence Is Defined Within Populations

“The study of the occurrence and distribution of health-related states or events in specified **populations**, including the study of the determinants influencing such states, and the application of this knowledge to control the health problems.”

Source: (Porta 2008)

Population

“The simplest definition of a **population** is a group of people who share characteristics or meet criteria that define membership in the population.”

Source: (Lash et al. 2021)

A Population Also Has a Time Period

“The simplest definition of a **population** is a group of people **during a defined time period** who share characteristics or meet criteria that define membership in the population.”

Source: (Lash et al. 2021)

Closed and Open Populations

CLOSED POPULATION

- After the population is defined, it does not add new members over time.
- It loses members only to death.

OPEN POPULATION

- It may gain members through birth or as people meet the membership definition.
- It loses members as people stop meeting the criteria or die.

Source: (Lash et al. 2021)

Closed vs. Open Populations: Calendar Time

Person	Enters	Observed through	Calendar-time experience
1	9/1	9/28	Four weeks
2	9/7	9/28	Three weeks
3	9/14	9/28	Two weeks
4	9/21	9/28	One week

Closed vs. Open Populations: Event Time

Aligning each person at entry changes the time scale from calendar time to event time.

Person	Event-time start	Observed through
1	0	4
2	0	3
3	0	2
4	0	1

What Happens Beyond the Observed Event Time?

Person	Observed event time	After observation ends
1	0-4	—
2	0-2	?
3	0-2	—
4	0-1	—

Examples of Closed and Open Populations

- Currently enrolled in Epidemiology III: **open population**
- Ever enrolled in Epidemiology III as of today: **closed population**
- Resident of Texas: **open population**
- People aged 65 or older as of today: **closed population**
- People aged 20 to 30 as of today: **open population**

Source and Target Populations

Source population: The population from which persons will be sampled and included in a measurement of disease frequency.

- The source population of the original Framingham Heart Study included men and women aged 30 to 62 who were residents of Framingham, Massachusetts, in 1948.

Target population: The group of people about which the scientific or public-health question asks and for whom the disease-frequency information will be relevant, we hope.

- Information about cardiac-disease risk factors from the Framingham Heart Study contributed to a nearly 75% decline in cardiovascular-disease mortality in most industrialized societies.

Sources: (Lash et al. 2021; Westreich 2019)

Study Population and Sample

Study population: The subset, up to a complete census, of the source population whose experience is included in a measurement of disease frequency.

- In the Framingham Heart Study, not everyone eligible was invited and not everyone invited agreed. Those who were invited, agreed, and were free of prevalent cardiovascular disease constituted the study population.

Sample: A selected subset of a population. A sample may be random or nonrandom and representative or nonrepresentative.

Sources: (Lash et al. 2021; Porta 2008)

Cohorts

- A **cohort** is a group of persons whose membership is defined permanently, or a population in which satisfying a set of defining events makes membership permanent.
- The term is often used as a synonym for “sample” in a cohort study.
- Every cohort member is enumerated: known or listed. This is not necessarily true of a closed population.

Source: (Lash et al. 2021)

Measures Arise From Observations and Experiments

Measures	Design types	Typical evidential position
Observations	Case descriptions; ecological; cross-sectional; case-control; cohort	Typically less convincing
Experiments	Non-randomized; randomized	Typically more convincing

Terminology Recap: Population Types

Term	Definition	Example
Population	A group of people during a defined time period who share characteristics or meet membership criteria.	The population of the United States.
Closed population	After the population is defined, it adds no new members and loses members only to death.	People who had a flu vaccine in 2022.
Open population	During a defined time frame, it gains members through birth or newly met criteria and/or loses members when criteria are no longer met or through loss to follow-up.	People who have the flu.

Terminology Recap: Source and Target Populations

Term	Definition	Example
Source population	The population from which persons will be sampled and included in a measurement of disease frequency.	Men and women aged 30 to 62 who lived in Framingham, Massachusetts, in 1948.
Target population	The group addressed by the scientific or public-health question and for whom the disease-frequency information will be relevant, we hope.	Industrialized societies to which Framingham findings about cardiac-disease risk factors were applied.

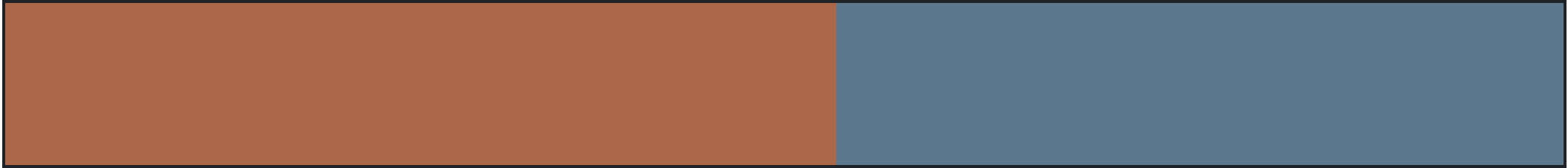
Terminology Recap: Study Population and Sample

Term	Definition	Example
Study population	The subset, up to a census, of the source population whose experience is included in the measurement.	Framingham invitees who agreed to participate and were free of prevalent cardiovascular disease.
Sample	A selected subset of a population; it may be random or nonrandom and representative or nonrepresentative.	Another, probably more common, term for study population.

Terminology Recap: Cohort

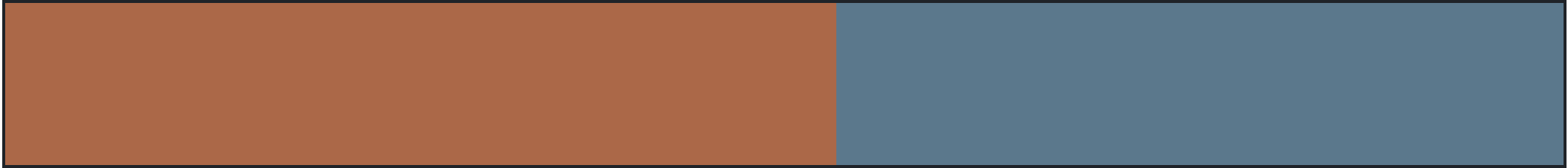
Term	Definition	Example
Cohort	A group whose membership is permanent once a defining set of events is satisfied. A cohort may be open or closed and is often used as a synonym for "sample" in a cohort study.	Similar to a population, except every cohort member is enumerated: known or listed.

Counts (n): Events in a Population



The population contains people with the event and people without the event.

Counts (n): Group the Event Categories



Grouping categories makes counting easier.

Counts (n): Reveal the Totals

EVENT: YES

53

EVENT: NO

47

Prevalence Describes Existing Conditions

- Prevalence provides information about the presence or absence of a condition of interest, such as a disease, exposure, or characteristic, in a population.
- It describes the extent to which the condition is present in a population during a given time frame.
- Asking about the prevalence of diabetes means asking how many, or more likely what proportion, of people currently living in the population have diabetes.

Source: (Westreich 2019)

Prevalence Count

- A count of people in a population with the condition of interest during a given time frame.
- As few as 0 people and as many as all people can be living with the condition.
- Range: 0 to infinity.

Source: (Westreich 2019)

Prevalence Proportion

- The proportion or percentage of the population that presently has the condition of interest or presently has a history of the condition during a specified period, such as a history of cancer in the past 10 years.
- Since none, some, or all population members can have the condition or a history of it, prevalence proportion ranges from 0 to 1, much like a probability.

Source: (Westreich 2019)

Prevalence Proportion: Formula

$$\text{Prevalence proportion} = \frac{\text{Count with the condition}}{\text{Living members of the population}}$$

Follow-Up Experience for 10 People

Person	At risk	Diseased	Deceased
1	0 months	Before follow-up–12	—
2	0–2	2–5	5–12
3	0–4	4–12	—
4	0–5	5–12	—
5	0–7	7–12	—
6	0–8	8–11	11–12
7	0–9	—	9–12
8	0–5	—	5–12
9	0–12	—	—
10	0–12	—	—

Month of follow-up. Source: (Lash et al. 2021)

Prevalence Proportion at Month 2

Person	Status at month 2
1	Diseased
2	Diseased
3–10	Living without disease

$$\frac{2}{10} = 0.20 = 20\%$$

Interpreting Prevalence Proportion

Person	Status at month 2
1	Diseased
2	Diseased
3–10	Living without disease

Among the members of the population, the prevalence of disease in month 2 was **0.20**.

Said another way, **20%** of the population had disease in month 2.

R



Source: R project logo

Point Prevalence and Period Prevalence

- **Point prevalence:** prevalence of a condition of interest at a single point in time.
- **Period prevalence:** prevalence of a condition of interest over a period of time.
- As the duration of the period shrinks, period prevalence converges to point prevalence.
- There is no hard-and-fast rule for what window is sufficiently short to count as point prevalence.

Source: (Westreich 2019)

Prevalence Odds

- Prevalence odds is a simple function of prevalence proportion, just as odds in general is a simple function of probability.
- It is typically reported for convenience or desirable statistical properties.
- Prevalence odds is prevalence proportion divided by 1 minus prevalence proportion.
- Range: 0 to infinity.

Source: (Westreich 2019)

Prevalence Odds: Formula

$$\text{Prevalence odds} = \frac{\text{Prevalence proportion}}{1 - \text{Prevalence proportion}}$$

Prevalence Odds at Month 2

$$\text{Prevalence odds} = \frac{0.20}{1 - 0.20} = 0.25$$

Interpreting Prevalence Odds

Among the members of the population, the odds of disease in month 2 were **0.25**.
For every person who had disease at month 2, there were four people who did not.

R



Source: R project logo

Incidence Describes New Occurrences

- Prevalence measures how many cases are present at a moment or over a period; incidence measures how many **new cases** arise over a specified period.
- Unlike prevalence, incidence measures occurrences or events: transitions from a condition being absent in an individual to the condition being present over a specified period.

Source: (Westreich 2019)

Incidence Proportion and Risk

- Incidence proportion and “risk” are sometimes used interchangeably in epidemiology.
- Risk can be ambiguous. Incidence proportion is less so.

Source: (Westreich 2019)

Incidence Count

- A count of new occurrences of a condition in a population at risk for the occurrence during a given time frame.
- Range: 0 to infinity.

Source: (Westreich 2019)

Incidence Proportion

- The proportion of the population that experiences a new occurrence of the condition among those at risk of experiencing a new occurrence during a given time frame.
- Like all proportions, it falls between 0 and 1.

Source: (Westreich 2019)

Incidence Proportion: Formula

$$\text{Incidence proportion} = \frac{\text{Count of new occurrences}}{\text{Population at risk}}$$

Follow-Up Experience and New Occurrences

Person	At risk	New disease	Deceased
1	0 months	Prevalent before follow-up	—
2	0–2	Month 2	5–12
3	0–4	Month 4	—
4	0–5	Month 5	—
5	0–7	Month 7	—
6	0–8	Month 8	11–12
7	0–9	—	9–12
8	0–5	—	5–12
9	0–12	—	—
10	0–12	—	—

Month of follow-up. Source: (Lash et al. 2021)

Incidence Proportion Over 12 Months

- 5 people developed disease.
- 9 people were at risk at baseline.
- Person 1 was already diseased and therefore not at risk for a new occurrence.

$$\frac{5}{9} = 0.56 = 56\%$$

Interpreting Incidence Proportion

Among the members of the population, the incidence of disease over 12 months of follow-up was **0.56**.

Fifty-six percent of the population developed disease over 12 months of follow-up.

Incidence Odds: Formula

$$\text{Incidence odds} = \frac{\text{Incidence proportion}}{1 - \text{Incidence proportion}}$$

Incidence Odds Over 12 Months

$$\text{Incidence odds} = \frac{0.56}{1 - 0.56} = 1.27$$

Interpreting Incidence Odds

Among the members of the population, the odds of incident disease over 12 months of follow-up were **1.27**.

For every 127 people who developed incident disease over 12 months of follow-up, 100 people did not.

Incidence Rate

- A measure of the average intensity at which an event occurs in people's experience over time.
- For a population followed from baseline, it is the number of cases divided by the person-time at risk accumulated by the population.
- Range: 0 to infinity.
- **Not a proportion.**
- The denominator is reciprocal time, usually person-time.
- The numeric value depends on the selected time unit and therefore has no interpretation by itself.

Source: (Westreich 2019)

Person-Time

Person-time is the amount of time observed for all people under study while they are at risk of experiencing an incident outcome.

Source: ([Westreich 2019](#))

Follow-Up Experience and Time at Risk

Person	Time at risk	Outcome after risk time
1	0 months	Prevalent disease
2	2 months	Disease, then death
3	4 months	Disease
4	5 months	Disease
5	7 months	Disease
6	8 months	Disease, then death
7	9 months	Death
8	5 months	Death
9	12 months	None observed
10	12 months	None observed

Source: (Lash et al. 2021)

Calculating Person-Time

$$\sum_{\text{people}} \text{Time at risk}$$

$$0 + 2 + 4 + 5 + 7 + 8 + 9 + 5 + 12 + 12 = 64 \text{ person-months}$$

Interpreting Person-Time

The population accumulated **64 person-months at risk** during 12 months of follow-up.

The same amount is **5.33 person-years at risk** during 12 months of follow-up.

Follow-Up Experience for the Incidence Rate

Person	Time at risk	New occurrence?
1	0 months	No; prevalent
2	2 months	Yes
3	4 months	Yes
4	5 months	Yes
5	7 months	Yes
6	8 months	Yes
7	9 months	No
8	5 months	No
9	12 months	No
10	12 months	No

Source: (Lash et al. 2021)

Incidence Rate: Calculation

$$\text{Incidence rate} = \frac{\text{Count of new occurrences}}{\text{Person-time at risk}}$$

$$= \frac{5}{64 \text{ person-months}} \approx 7.8 \text{ per 100 person-months}$$

Interpreting the Incidence Rate

The incidence rate of disease among the members of the population was **5 cases per 64 person-months** during 12 months of follow-up.

Equivalently, the incidence rate was **7.8 cases per 100 person-months** during 12 months of follow-up.

Odds and Proportions

$$\text{Odds} = \frac{p}{1 - p}$$

<i>p</i>	Odds
0.01	0.010
0.10	0.111
0.20	0.250
0.50	1.000
0.80	4.000
0.90	9.000
0.99	99.000

Source: (Westreich 2019)

References

- Lash, Timothy L., Tyler J. VanderWeele, Sebastien Haneuse, and Kenneth J. Rothman. 2021. *Modern Epidemiology*. 4th ed. Wolters Kluwer.
- Porta, Miquel, ed. 2008. *A Dictionary of Epidemiology*. Oxford University Press.
- Westreich, Daniel. 2019. *Epidemiology by Design: A Causal Approach to the Health Sciences*. Oxford University Press.